

# ENVS 193DS Homework 3

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## Set up

```
# loaded in packages
library(tidyverse)
library(janitor)
library(here)
library(readxl)

# read in data and saved as object called `kelp`
kelp <- readr::read_csv(here::here("data","temp-kelp.csv"))
# read in my data and saved as object called 'rxn-time'
rxn-time <- readr::read_csv(here::here("data","rxn-time.csv"))
```

## Problem 1. Giant kelp fronds

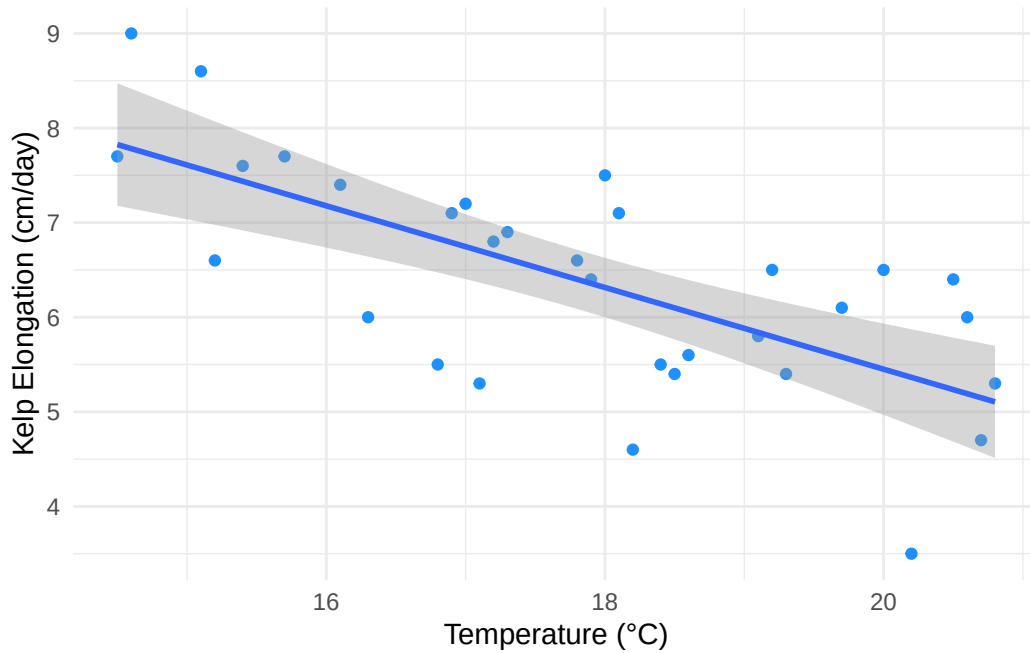
### a. An appropriate test

Pearson's correlation and ordinary least squares (OLS) can be used to test the strength of the relationship between temperature and giant kelp frond elongation rate. Whereas Pearson's  $r$  would indicate the strength and direction of linear association between the two variables (temperature and kelp elongation rate), the OLS slope would predict changes in the response variable (kelp elongation rate) based on the value of the predictor variable (temperature) and can account for multiple variables.

### b. Create a visualization

```
# base layer: ggplot
ggplot(kelp,
  # x-axis: temperature
  # y-axis: kelp elongation
  aes(x = temp_c,
    y = kelp_elong)) +
# added raw data points with custom color
```

```
geom_point(color = "dodgerblue") +
# fitted a linear model
geom_smooth(method = "lm", se = TRUE) +
# relabeled x and y axes
labs(x = "Temperature (°C)",
     y = "Kelp Elongation (cm/day)") +
# changed default theme
theme_minimal()
```



### c. Check your assumptions and run your test

#### Check your assumptions

```
# fitted model
kelp_model <- lm(
  kelp_elong ~ temp_c,
  data = kelp)

# model information
summary(kelp_model)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

|  | Min     | 1Q      | Median | 3Q     | Max    |
|--|---------|---------|--------|--------|--------|
|  | -1.8643 | -0.5017 | 0.1790 | 0.5659 | 1.2177 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 14.08645 | 1.49219    | 9.440   | 1.72e-10 *** |
| temp_c      | -0.43179 | 0.08321    | -5.189  | 1.37e-05 *** |

---

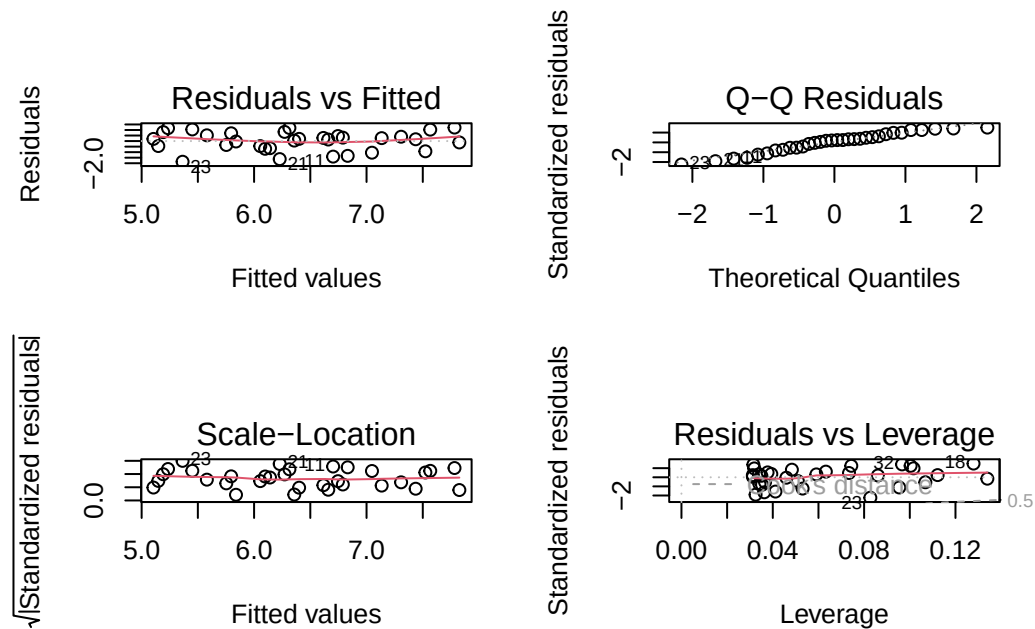
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

```
# base R residuals  
par(mfrow = c(2,2))  
plot(kelp_model)
```



I checked that there was a linear relationship between variables (plotted data appears to adhere to a linear trend). I also checked that errors were independent of each other (plot residuals appear randomly scattered), homoscedastic (residuals vs fitted plot points scattered evenly), and normally distributed (QQ plot points closely follow the trend line).

### Run test

```
# Pearson's correlation test
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "pearson")
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

### d. Results communication

I used a Pearson's correlation test to evaluate the strength of the relationship between temperature and giant kelp frond elongation rate because it tests the strength and direction of the relationship between two variables (in this case, temperature and kelp frond elongation rate). We found a moderate negative relationship between temperature and kelp frond elongation rate (Pearson's  $r = -0.69$ ,  $t(30) = -5.2$ ,  $p = 0.00001$ ,  $\alpha = 0.05$ ).

### e. Test implications

The results of this test imply that increasing ocean temperatures are moderately associated with decreasing giant kelp frond elongation rates. Climate change has been altering the physical and chemical properties of oceans globally; since kelp forests provide various ecosystem services to humans and the world at large, it may be worthwhile to further investigate the impacts of rising temperatures on their growth.

## f. Double check your own work

```
# fitted model  
summary(kelp_model)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -1.8643 | -0.5017 | 0.1790 | 0.5659 | 1.2177 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 14.08645 | 1.49219    | 9.440   | 1.72e-10 *** |
| temp_c      | -0.43179 | 0.08321    | -5.189  | 1.37e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

```
# slope confidence interval  
confint(kelp_model)
```

|             | 2.5 %      | 97.5 %     |
|-------------|------------|------------|
| (Intercept) | 11.0389930 | 17.1338986 |
| temp_c      | -0.6017337 | -0.2618482 |

The OLS slope tests the predicted change in kelp elongation as it relates to temperature and fits the model accordingly. The slope is negative and significant, meaning that both tests lead to the same decision: there is a significant, negative linear relationship between temperature and kelp frond elongation rates.

## Problem 2. Personal data

### a. Updating your visualizations

#### Cleaning data

```
# created a new object from rxntime
rxntime_type <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_type, reaction_time_avg_ms) |>
  # removed missing values
  drop_na(exercise_type, reaction_time_avg_ms) |>
  # capitalized existing exercise type categories
  mutate(exercise_type = case_when(
    exercise_type == "biking" ~ "Biking",
    exercise_type == "volleyball" ~ "Volleyball",
    exercise_type == "walking" ~ "Walking"
  ))
```

```
# created a new object from rxntime
rxntime_clean <- rxntime |>
  # cleaned column names
  clean_names() |>
  # selected columns of interest
  select(exercise_duration_min, reaction_time_avg_ms) |>
  # removed missing values
  drop_na(exercise_duration_min, reaction_time_avg_ms)
```

Figure 1. Exercise type and reaction time

```
# base layer: ggplot
ggplot(data = rxntime_type,
  # x-axis: exercise type
  # y-axis: average reaction time in milliseconds
  # color by exercise type
  mapping = aes(x = "Exercise Type",
    y = "Average Reaction Time (ms)",
```

```

        color = exercise_type)) +
# first layer: data points
geom_point(data = rxntime_type,
           aes(x = exercise_type,
              y = reaction_time_avg_ms,
              size = 3)) +
# changed theme
theme_minimal() +
# removed legend
theme(legend.position = "none",
      panel.background = element_blank(), # removing panel background
      panel.grid = element_blank(), # removing panel grid
      axis.ticks = element_blank() # removing axis ticks)
) +
# relabeled x- and y- axes, adding title
labs(title = "Exercise Type and Reaction Time",
      subtitle = "May 29, 2026",
      x = "Exercise Type",
      y = "Average Reaction Time (ms)") +
# custom colors
scale_color_manual(values =
                   c("Biking" = "magenta",
                     "Volleyball" = "purple3",
                     "Walking" = "blue"))

```



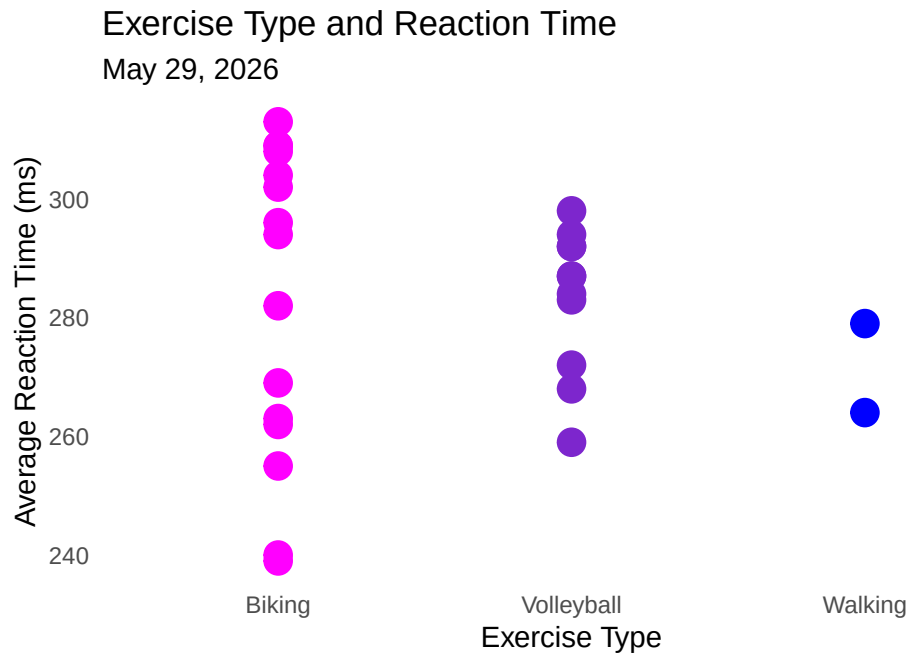


Figure 2. Exercise duration and reaction time

```
# base layer: ggplot
ggplot(data = rxntime_clean,
  # x-axis: exercise duration in minutes
  # y-axis: average reaction time in milliseconds
  # color
  mapping = aes(x = "Exercise Type",
    y = "Average Reaction Time (ms)")) +
# first layer: data points
geom_point(data = rxntime_clean,
  aes(x = exercise_duration_min,
    y = reaction_time_avg_ms,
    size = 3,
    color = "hotpink")) +
# changed theme
theme_minimal() +
# removed legend
theme(legend.position = "none",
  panel.background = element_blank(), # removing panel background
  panel.grid = element_blank(), # removing panel grid
```

```

    axis.ticks = element_blank(), # removing axis ticks
  ) +
  # relabeled x- and y- axes, adding title
  labs(title = "Exercise Duration and Reaction Time",
        subtitle = "May 29, 2026",
        x = "Exercise Duration (min)",
        y = "Average Reaction Time (ms)") +
  # custom colors
  scale_color_manual(values = "hotpink")

```



## b. Captions

**Figure 1. Mean reaction times across biking, volleyball, and walking exercises.** Points represent observations of mean reaction time (ms) after different kinds of exercise (pink: Biking, purple: volleyball, blue: walking). Data from my personal observations. Data accessed 2026-05-29.

**Figure 2. Mean reaction times and exercise duration.** Points represent observations of mean reaction time (ms) in relation to exercise duration (min). Data from my personal observations. Data accessed 2026-05-29.

### Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data.

Taking inspiration from the Dear Data project, I could make a visual where each observation is represented by an icon (a stick figure for walking, a volleyball, and a bike tire for biking). Additional doodles such as a water droplet or a star can indicate whether water was consumed throughout exercise, or if an exercise exceeded a certain amount of time. Colors can be used to indicate levels of social interaction.

b. Create a sketch (on paper) of your idea.

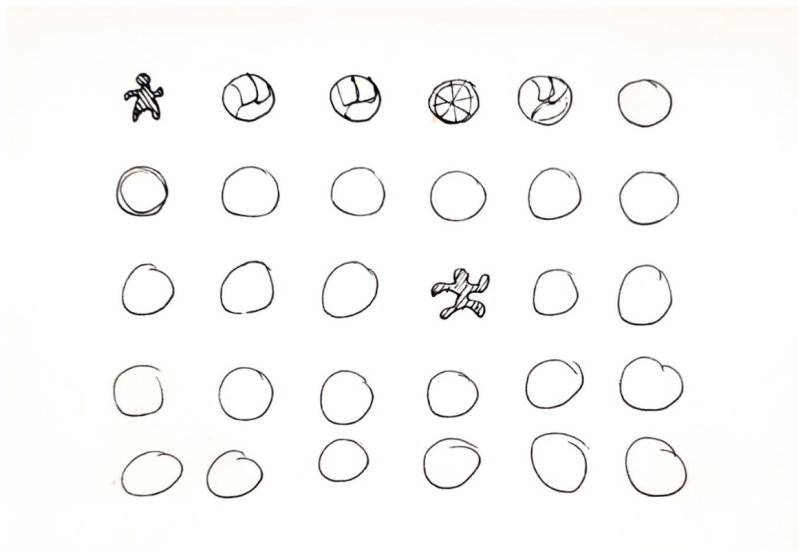


Figure 1

c. Make a draft of your visualization.

d. Write an artist statement.

This is a visualization of the exercises I have done throughout the spring quarter. The icons arranged in an array of thirty observations each represent a type of exercise (walking, biking, or volleyball) while indicating factors such as socialization with other people and hydration. I was inspired most by the Dear Data project; especially by how colors and shapes were used not only to embellish the visualization but to tell a story. I created this draft with a sheet of A4 paper, a pen, and a couple highlighters. It is hand-drawn.

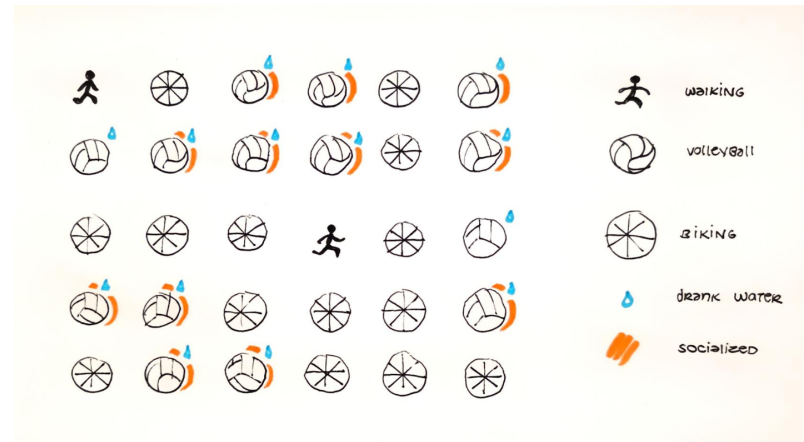


Figure 2

**e. Prep your materials to share in class.**

View presentation slides [here](#).

## Problem 4. Statistical critique

**a. Revisit and summarize**

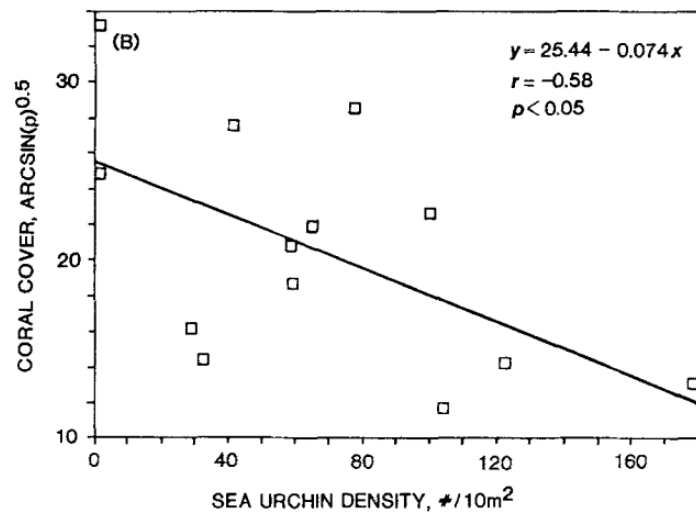
Pearson's R tests were used in this experiment to determine whether the relationship between coral cover and sea urchin density across all six studied reefs (Malindi, Watamu, Vipingo, Kanamai, Bamburi, and Diani) was significant.

**b. Visual clarity**

The statistics are clearly represented in this linear model; the x- and y-axes are in logical positions while the line of best fit, the r-value, p-value, model predictions (via the line of best fit), and underlying data points are present.

**c. Aesthetic clarity**

The data:ink ratio appears to be low. Visual clutter of the data is kept to a minimum by only including values that are necessary to understand it. This linear regression model is, structurally, quite neat as it is used to display the relationship between two variables. There is a slight sense of compaction, but that can be attributed to the densely-packed nature of the paper itself.



**Fig. 7.** (A) topographic complexity and (B) arcsine transformed coral cover plotted against the total sea urchin density

Figure 3

#### d. Recommendations

I would try to further visually separate the variables (coral cover and sea urchin density) from their units. Currently, they are separated by a comma, which may be easily overlooked by quick eyes and potentially confuse the reader. If this paper were to be made available not just in black print, I think that the raw data and trend line would look aesthetically pleasing in different colors (and allow them to stand out visually from the rest of the figure).